

Problem

A band-limited voltage signal is found to have the complex Fourier coefficients presented in the table below. Calculate the average power that the signal would supply a 4-Ω resistor.

$n\omega_0$	$ c_n $	θ_n
0	10.0	0°
ω	8.5	15°
2ω	4.2	30°
3ω	2.1	45°
4ω	0.5	60°
5ω	0.2	75°

Step-by-step solution

Step 1 of 2

$$P = \frac{V_{DC}^2}{R} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{V_n^2}{R}$$

Assuming V is an amplitude - phase form of Fourier series byt,

$$|A_n| = 2|G|, \quad C_o = a_o$$

$$|A_n|^2 = 4|C_n|^2$$

Hence,

$$P = \frac{C_o^2}{R} + 2 \sum_{n=1}^{\infty} \frac{C_n^2}{R}$$

Step 2 of 2

Alternatively

$$P = \frac{V_{rms}^2}{R}$$

Where

$$\begin{aligned} V_{rms}^2 &= a_o^2 + \frac{1}{2} \sum_{n=1}^{\infty} A_n^2 = C_o^2 + 2 \sum_{n=1}^{\infty} C_n^2 = \sum_{n=-\infty}^{\infty} C_n^2 \\ &= 10^2 + 2(8.5^2 + 4.2^2 + 2.1^2 + 0.5^2 + 0.2^2) \\ &= 10^2 + 2 \times 94.57 = 289.14 \end{aligned}$$

$$P = \frac{289.14}{4} = 723 \text{ Watt}$$